

15 The Chromosomal Basis of Inheritance 294

CONCEPT 15.1 Mendelian inheritance has its physical basis in the behavior of chromosomes 295

- Morgan's Choice of Experimental Organism 295
- Correlating Behavior of a Gene's Alleles with Behavior of a Chromosome Pair: *Scientific Inquiry* 295

CONCEPT 15.2 Sex-linked genes exhibit unique patterns of inheritance 298

- The Chromosomal Basis of Sex 298
- Inheritance of X-Linked Genes 299
- X Inactivation in Female Mammals 300

CONCEPT 15.3 Linked genes tend to be inherited together because they are located near each other on the same chromosome 301

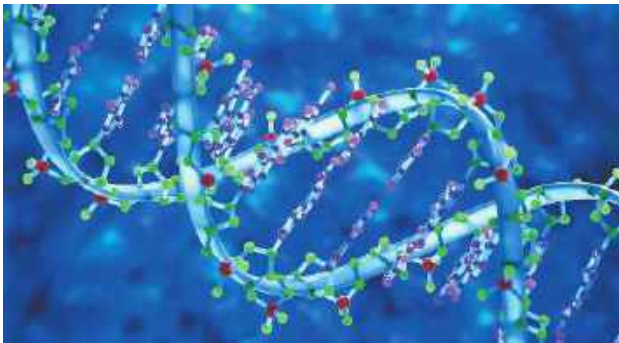
- How Linkage Affects Inheritance 301
- Genetic Recombination and Linkage 302
- Mapping the Distance Between Genes Using Recombination Data: *Scientific Inquiry* 305

CONCEPT 15.4 Alterations of chromosome number or structure cause some genetic disorders 306

- Abnormal Chromosome Number 307
- Alterations of Chromosome Structure 307
- Human Conditions Due to Chromosomal Alterations 308

CONCEPT 15.5 Some inheritance patterns are exceptions to standard Mendelian inheritance 310

- Genomic Imprinting 310
- Inheritance of Organelle Genes 311



16 The Molecular Basis of Inheritance 314

CONCEPT 16.1 DNA is the genetic material 315

- The Search for the Genetic Material: *Scientific Inquiry* 315
- Building a Structural Model of DNA 317

CONCEPT 16.2 Many proteins work together in DNA replication and repair 320

- The Basic Principle: Base Pairing to a Template Strand 321
- DNA Replication: *A Closer Look* 322
- Proofreading and Repairing DNA 327
- Evolutionary Significance of Altered DNA Nucleotides 328
- Replicating the Ends of DNA Molecules 328

CONCEPT 16.3 A chromosome consists of a DNA molecule packed together with proteins 330

17 Gene Expression: From Gene to Protein 335

CONCEPT 17.1 Genes specify proteins via transcription and translation 336

- Evidence from Studying Metabolic Defects 336
- Basic Principles of Transcription and Translation 337
- The Genetic Code 340

CONCEPT 17.2 Transcription is the DNA-directed synthesis of RNA: *A Closer Look* 342

- Molecular Components of Transcription 342
- Synthesis of an RNA Transcript 342

CONCEPT 17.3 Eukaryotic cells modify RNA after transcription 345

- Alteration of mRNA Ends 345
- Split Genes and RNA Splicing 345

CONCEPT 17.4 Translation is the RNA-directed synthesis of a polypeptide: *A Closer Look* 347

- Molecular Components of Translation 348
- Building a Polypeptide 350
- Completing and Targeting the Functional Protein 352
- Making Multiple Polypeptides in Bacteria and Eukaryotes 355

CONCEPT 17.5 Mutations of one or a few nucleotides can affect protein structure and function 357

- Types of Small-Scale Mutations 357
- New Mutations and Mutagens 360
- Using CRISPR to Edit Genes and Correct Disease-Causing Mutations 360
- What Is a Gene? *Revisiting the Question* 361

18 Regulation of Gene Expression 365

CONCEPT 18.1 Bacteria often respond to environmental change by regulating transcription 366

- Operons: The Basic Concept 366
- Repressible and Inducible Operons: Two Types of Negative Gene Regulation 368
- Positive Gene Regulation 369

CONCEPT 18.2 Eukaryotic gene expression is regulated at many stages 370

- Differential Gene Expression 370
- Regulation of Chromatin Structure 371
- Regulation of Transcription Initiation 373
- Mechanisms of Post-transcriptional Regulation 377

CONCEPT 18.3 Noncoding RNAs play multiple roles in controlling gene expression 379

- Effects on mRNAs by MicroRNAs and Small Interfering RNAs 379
- Chromatin Remodeling and Effects on Transcription by ncRNAs 380

CONCEPT 18.4 A program of differential gene expression leads to the different cell types in a multicellular organism 381

- A Genetic Program for Embryonic Development 381
- Cytoplasmic Determinants and Inductive Signals 382
- Sequential Regulation of Gene Expression During Cellular Differentiation 383
- Pattern Formation: Setting Up the Body Plan 384